Your Map Is Its Own Calibration Set: Label-Free, Shift-Robust Rejection of False Matches in Visual Place Recognition

Abstract

Visual place recognition (VPR) returns a best match for every query, yet a robot must decide whether to **trust** it: accepting a match for a place that is not in the map — or one that merely looks like it — corrupts pose graphs and relocalisation. Existing acceptance rules (similarity thresholds, learned integrity monitors, conformal calibration) all need labelled query–map pairs, and we show that their false-accept guarantees break as soon as the appearance condition (night, rain, snow, season) differs between calibration and deployment. We observe that VPR offers labelled negatives for free: **the map itself**. Holding out each spatial block of the map turns every map image into a query that is off-map **by construction**. Calibrating an acceptance statistic against these map pseudo-queries yields split-conformal p-values without any human or deployment labels. To make this calibration transfer across appearance conditions we use statistics that are invariant to per-query affine changes of the similarity scores, computed against spatially distant map regions on hubness-reduced (map-centred) descriptors, combine them with a self-calibrated minimum-p rule, and add an aliasing guard based on the gap to the best distant region. We prove validity under exchangeability, conservativeness whenever appearance change only makes unrelated queries less map-like, and FDR control with Benjamini–Hochberg. On RobotCar-Seasons (official reference map and thresholds), Extended-CMU-Seasons (eleven official slices merged into one 2,418-image city map, 21k queries, 11 dates) and Nordland, with six descriptors from NetVLAD to MegaLoc, the method keeps the off-map false-accept rate at or below the target in all 138 descriptor × condition cells and rejects nearly all perceptual-aliasing errors (median acceptance 0.8%), while label-calibrated baselines exceed the target in 25–64 of the 138 cells — leave-one-condition-out conformal calibration fails in every one of the 18 descriptor–dataset settings — and accept 17–44% of aliasing errors. Without the aliasing guard, its average power equals that of an oracle that has labels for every deployment condition (0.718 vs 0.721).

1. Introduction

Modern VPR descriptors are remarkably robust: on RobotCar-Seasons a frozen SALAD descriptor retrieves a reference image within 5 m and 10° for about 90% of queries even at night. Recall, however, is not what breaks a deployed system. A VPR module always returns a top-1 match, and the downstream consumer — loop closure in SLAM, relocalisation, map merging — needs to know **whether to accept it**. Two failure modes dominate. First, the query may come from a place that is **not in the map** (an unmapped street, a detour, a changed environment). Second, the query may be in the map but be matched to a **perceptual alias**, a different place that looks the same. A single accepted false match can corrupt a pose graph irreversibly; a rejected true match merely delays localisation. The decision therefore calls for an explicit, controllable false-accept rate.

The standard recipe is to calibrate a threshold — on raw similarity, on a hand-crafted score such as the top-1/top-2 ratio, or on the output of a learned integrity monitor — using queries whose correctness is known, possibly with a conformal procedure that turns the calibration set into a finite-sample guarantee [SAFEVPR, Quantile Transfer]. These guarantees rest on exchangeability between calibration and deployment queries. VPR is the textbook case in which this assumption fails: the whole point of the task is that deployment happens under different illumination, weather and season than any data one could label in advance. We confirm this empirically (§6): thresholds calibrated with labels on benign conditions exceed a 5% false-accept target by up to 4× at night, and even calibrating on all **other** labelled conditions (the protocol of SAFEVPR) leaves violations in a third of the deployment conditions on Extended-CMU.

We make three observations that, together, remove the need for labels.

1. **The map provides labelled negatives for free.** A VPR map is a set of images with known positions. Removing a spatial block from the map and querying it with one of its own images produces a query that is off-map **by construction**. Repeating this for all blocks yields thousands of null examples in the map's own environment — exactly the distribution a deployment-time off-map query is drawn from, up to the appearance condition.

2. **Appearance change acts mostly as an affine distortion of similarity scores**, and appearance change can only make an unrelated query **less** similar to the map than the map is to itself. Statistics that are invariant to per-query affine changes therefore transfer from the map's condition to any deployment condition, and the residual shift is in the safe (conservative) direction.
3. **Map geometry separates the two failure modes.** Scoring the top-1 against the best matches of **spatially distant** map regions (rather than its own neighbours, which are visually redundant) gives a statistic for off-map detection; the gap to the single best distant region exposes aliasing, where the true place is a strong distant competitor.

We call the result **self-calibrating maps (SelfCal)**. It is training-free, label-free, descriptor-agnostic, costs one offline pass of self-retrieval over the map and a sort per query, and provides a p-value per query that can be used with a fixed level, with Benjamini–Hochberg for batches, or turned into a calibrated confidence via the two-groups model — all without labels.

Contributions. (i) We show on three benchmarks that label-calibrated acceptance (threshold, learned and conformal) loses its false-accept guarantee under appearance shift, and that the classical parametric nulls (Gaussian z-norm, EVT spacing tests) are invalid or degraded by hubness. (ii) We propose SelfCal: map pseudo-queries via spatial hold-out as a conformal null sample, affine-invariant far-region statistics on map-centred descriptors, a self-calibrated combination, and an aliasing guard. (iii) We give validity, dominance-based robustness, excess-risk and FDR results. (iv) We evaluate on RobotCar-Seasons, Extended-CMU-Seasons and Nordland with six descriptors and 23 deployment conditions, with full ablations; code and descriptor dumps are released.

2. Related work

Uncertainty and introspection for VPR. Retrieval-based scores — the L2 distance to the nearest reference and the perceptual-aliasing (PA) ratio of the first two distances [Hausler'21] — remain strong baselines [Zaffar'24]. Zaffar et al. [CVPR'24] compare them with learned aleatoric uncertainty (BTL [Warburg'21], STUN [Cai'22]) and geometric verification, and propose SUE, the spatial spread of the top-K reference poses. Miller et al. [RA-L'26] introduce training-free score-distribution metrics (ratio spread, similarity distribution). KappaPlace [2026] learns a von Mises–Fisher concentration and reports calibration error. Integrity monitors learn to predict whether a match is within tolerance from statistics of the distance vector, with SVMs [Carson'22] or MLPs [Claxton'24]; Hussaini et al. predict when sequence matching helps [2025]. All of these produce **scores**; turning a score into an accept/reject decision with a known error rate requires labelled calibration data from the deployment conditions. SelfCal is complementary: it can calibrate any score, and it shows which scores transfer across conditions.

Thresholds and operating points. Schubert et al. [ICRA'21] model the similarity distribution as a robust normal and threshold a high quantile; Vysotska et al. [ICRA'25] adapt thresholds for sequences; Trinh & Lyons [2025] derive per-place thresholds from negative statistics of the database; Quantile Transfer [IROS'26] transfers a threshold calibrated on a labelled traversal through quantiles of the deployment similarity distribution. None provides a guarantee under appearance shift; parametric nulls are invalid in our experiments and raw-score database negatives are conservative but lose power under shift (§6.4).

Conformal prediction for robotics and VPR. Split conformal prediction and its risk-control extensions provide finite-sample guarantees under exchangeability [Vovk'05, Angelopoulos'21, Bates'23]; Barber et al. [2023] bound the coverage gap under distribution shift. SAFEVPR [Sier'26] combines a DINOv2 patch-matching verifier with Mondrian Learn-then-Test and reports empirical FDR validity across conditions when calibrating on labelled data of the other conditions; Route-MHT [2026] learns a map-out guardrail for thermal VPR. Our null sample comes from the map rather than from labelled queries, and our robustness argument is a stochastic-dominance property of the map's own condition.

Using the map at test time. The reference map with its poses is available before deployment in most VPR applications; it has been used to fine-tune descriptors (RSF [Zaffar'25]), to choose map density from two reference traversals [Hussaini'26], and to estimate spatial uncertainty (SUE). We use it as a **calibration set**.

Hubness. High-dimensional nearest-neighbour retrieval produces hubs, points that are close to many queries [Radovanović'10]; centring [Suzuki'13] and local scaling (CSLS [Conneau'18]) reduce it. We find that hubness is what breaks extreme-value nulls in VPR and that map centring improves the power of the invariant statistic.

3. Problem

A map $\mathcal{M}={(f_j,x_j)}_{j=1}^N$ holds L2-normalised global descriptors f_j and positions x_j (metres; frame index on Nordland). A query q with descriptor f_q yields similarities $s_j=\langle f_q,f_j\rangle$ and the retrieval $j^*=\arg\max_j s_j$. An **acceptance test** maps the similarity vector (and map positions) to accept/reject. We distinguish:

- **Off-map false accept:** q has no map position within δ (40 m) and is accepted. The false-accept rate in deployment condition c is $\mathrm{FAR}_c(\alpha)=P(\text{accept}\mid q\text{ off-map}, c)$.
- **Aliasing accept:** q is in the map, but x_{j^*} is farther than ρ (25 m) from its true position.
- **True accept (power):** q is in the map and the top-1 satisfies the localisation tolerance (5 m and 10° on RobotCar/CMU; 2 frames on Nordland).

The goal is $\mathrm{FAR}_c(\alpha)\le\alpha$ for every condition c, few aliasing accepts and high power, **without labelled queries**.

4. Method: self-calibrating maps

Map pseudo-queries. Partition map positions into square blocks of side L (300 m; 200 frames on Nordland). For a map item i in block b whose distance to $\mathcal{M}\setminus B_b$ exceeds δ, compute the acceptance statistic of f_i against the reduced map, $T^0_i=T(f_i;\mathcal{M}\setminus B_b)$. Item i is an off-map query by construction. Using every second map image gives 757 (CMU) to 2,391 (RobotCar) null statistics.

Far-region statistic. Hash map positions into cells of side 10 m, take the region maxima $Z_c=\max_{j\in c}s_j$, and keep the cells farther than ρ from x_{j^*}. **With mean μ_{far} and standard deviation σ_{far} of these far-region maxima,**
$$z_{far}(q)=\frac{s_{j^*}-\mu_{far}}{\sigma_{far}},\qquad z_{gap}(q)=\frac{s_{j^*}-\max_{c\,\text{far}}Z_c}{\sigma_{far}}.$$ **Both are invariant to $s\mapsto as+b$ ($a>0$). Excluding the neighbourhood of j^*** prevents the visually redundant neighbours of a true match from being counted as competitors; using region maxima makes the competitors comparable.

Hubness reduction. Descriptors are centred on the map mean and re-normalised (map only, no labels).

Conformal p-value. $p_T(q)=\big(1+\#\{i: T^0_i\ge T(q)\}\big)/(1+n)$.

Self-calibrated combination. No single statistic is best everywhere (raw max-similarity is strongest on CMU, the invariant z_{far} on Nordland). We take $m(q)=\min\{p_{\max s}(q),p_{z_{far}}(q)\}$ and re-calibrate it against the leave-one-out values $m(i)$ of the null items: $p_{comb}(q)=(1+\#\{i:m(i)\le m(q)\})/(1+n)$.

Aliasing guard (SelfCal-AA). Accept iff $\max\{p_{comb}(q),\,p_{z_{gap}}(q)\}\le\alpha$. In an aliasing failure the true place is a strong far competitor, so z_{gap} is small.

Batches. With p-values of a deployment batch, Benjamini–Hochberg controls the false-discovery rate, and the two-groups model (Storey's $\hat\pi_0$, Grenander density estimate) turns p into a calibrated probability of a true match, $1-\widehat{\mathrm{lfdr}}(p)$ — again without labels.

Cost. Offline: one self-retrieval pass over the map with block hold-out — 24 s on 4 CPU cores for the 6,954-image RobotCar map (8,448-d SALAD). Online: the similarity vector (needed for retrieval anyway), its centred counterpart, cell maxima (one `reduceat`) and three binary searches — about 10 ms per query on CPU in NumPy, both similarity products included.

5. Guarantees

T1 (validity). If, for an off-map query, $T(q)$ and the null statistics ${T^0_i}$ are exchangeable, then $P(p_T(q)\le\alpha)\le\alpha$. **Standard split-conformal argument.**

T2 (dominance ⇒ conservative). If in condition c the off-map statistic is stochastically dominated by the map null, $P(T^c\ge t)\le P(T^0\ge t)\ \forall t$, then $\mathrm{FAR}_c(\alpha)\le\alpha$. Map pseudo-queries are images of the map's own traversal: appearance change can only make an unrelated query **less** map-like, so calibrating on the clean map errs on the safe side. Fig. 1 verifies dominance for all nine RobotCar conditions.

T3 (excess risk). In general FAR**$c(\alpha)-\alpha\le\Delta_c^+:=\sup_t[P(T^c\ge t)-P(T^0\ge t)]^+$**, the one-sided Kolmogorov distance. Affine invariance removes the dominant per-query distortion, keeping Δ_c^+ near zero while keeping the null tight (power).

T4 (combination). If the vector of statistics is exchangeable with the null vectors, the re-calibrated minimum is a valid conformal p-value (the leave-one-out ranks are computed exactly as for the query).

T5 (FDR). Conformal p-values that share one calibration set are PRDS [Bates'23]; hence Benjamini–Hochberg at level q controls the off-map FDR at $\pi_0 q$. Aliasing control is empirical: an in-map aliasing failure differs from its off-map counterpart by an extra strong far competitor, which can only decrease z_{far} and z_{gap}.

6. Experiments

6.1 Setup

Datasets. RobotCar-Seasons v2 [Sattler'18]: the map is the official `overcast-reference` rear-camera traversal (6,954 images, ~1.3 × 1.1 km) with poses from the official reconstruction (`all.nvm`); queries are the 1,906 rear images of the public training split in nine conditions (dawn, dusk, night, night-rain, overcast-summer/winter, rain, snow, sun), whose 6-DoF poses are released (test poses are withheld by the benchmark). A query is correct if the top-1 reference is within the coarsest official threshold (5 m, 10°); we also report the finer official thresholds. **Extended-CMU-Seasons** [Toft'20]: the eleven slices with released query poses (6–8, 10–13, 21–24) share one world frame and are consecutive segments, so we merge their official databases into one 2,418-image city map; queries are the 21,407 camera-0 images of 11 dates (Sep 2010 – Jul 2011), correct at (5 m, 10°). **Nordland** [Sünderhauf'13]: 4,500 frame-aligned places per season (uniform subsample of the 35,768-frame route); summer is the map (winter in §6.6), spring/fall/winter are queries, correct within 2 places.

Off-map protocol. We cannot rely on a few hand-picked unmapped streets, so we hold out the map in turn: the map is partitioned into 300 m blocks (200 places on Nordland); for each block, the block is removed and its queries farther than 40 m (10 places) from the remaining map are evaluated as **off-map**. This gives 1,386 (RobotCar), 14,117 (CMU) and 12,180 (Nordland) off-map query instances spread over all conditions. In-map queries (within 5 m of a map image) are evaluated against the full map; in-map queries whose top-1 is farther than 25 m (5 places) are **aliasing** errors. The method's own null construction (map blocks, pseudo-queries) uses only the map and never sees queries.

Descriptors. Frozen, off-the-shelf: NetVLAD (VGG16, Pitts30k, 4,096-d) [Arandjelović'16], CosPlace (ResNet-50, 2,048-d) [Berton'22], MixVPR (ResNet-50, 4,096-d) [Ali-bey'23], DINOv2 ViT-B/14 CLS+mean (1,536-d) [Oquab'24], SALAD (8,448-d) [Izquierdo'24] and MegaLoc (8,448-d) [Berton'25]. All descriptors are extracted once at 322 × 322.

Baselines. (a) **Threshold, benign→all**: split-conformal threshold on the max-similarity of off-map queries of the two most benign conditions (RobotCar: overcast-summer/winter; CMU: the two dates with the highest recall; Nordland: fall), applied to all conditions; (b) the same with aliasing errors added to the negatives; (c) **conformal LOCO**, the SAFEVPR protocol: calibrate on the labelled off-map queries of all **other** conditions and test on the held-out one; (d) **Quantile Transfer** [Rajani'26]: the benign threshold mapped through the quantile function of each deployment batch; (e) a learned logistic calibrator on margin, max-similarity and entropy (CRSG) trained on benign conditions; (f) the PA ratio [Hausler'21]; (g) parametric nulls: Gaussian z-norm with Bonferroni and the closed-form EVT spacing test; (h) an **oracle** that calibrates max-similarity with the labels of every deployment condition.

Metrics. FAR on off-map queries per condition (target α), the number of conditions whose FAR is significantly above α (one-sided binomial test, Bonferroni over conditions), TPR on correct in-map queries, and the acceptance rate of aliasing errors.

6.2 Main result

Table 1 reports α = 0.05 for SALAD. Every label-calibrated method exceeds the target in at least one realistic dataset — the benign→all threshold in 5/9 RobotCar conditions (worst FAR 0.21 at night-rain) and 4/11 CMU dates; calibrating on all other labelled conditions (SAFEVPR protocol) still fails in 1/9, 4/11 and 2/3 conditions — and all of them accept a third to four fifths of the aliasing errors. Parametric nulls are either invalid (Gaussian z-norm: FAR 0.13–0.35) or anti-conservative under hubness (EVT spacing: night-rain 0.109). SelfCal-AA, which uses no labels, has **no significant violation in any of the 23 conditions**, accepts 0.0%, 1.5% and 5.7% of aliasing errors, and its TPR (0.988 / 0.853 / 0.754) is close to the oracle that holds labels for every condition (0.995 / 0.992 / 0.780). Without the aliasing guard, SelfCal reaches oracle power on RobotCar and CMU (0.994 / 0.956) while still controlling the off-map FAR. Fig. 2 shows the same picture for $\alpha \in [0.005, 0.2]$.

Table 1 (α = 0.05). SALAD. Cell: off-map FAR / #conditions significantly above α (binomial, Bonferroni) / TPR / aliasing-accept rate

method	labels	RobotCar-Seasons (9 cond.)	Extended-CMU (11 dates)	Nordland (3 seasons)
SelfCal-AA (ours)	none	0.004 / 0/9 / 0.988 / 0.00	0.002 / 0/11 / 0.853 / 0.01	0.008 / 0/3 / 0.754 / 0.06
SelfCal (ours, no alias guard)	none	0.004 / 0/9 / 0.994 / 0.01	0.009 / 0/11 / 0.956 / 0.26	0.031 / 0/3 / 0.788 / 0.52
Threshold, benign→all	benign	0.109 / 5/9 / 0.997 / 0.50	0.052 / 4/11 / 0.985 / 0.78	0.037 / 0/3 / 0.630 / 0.49
Threshold, benign, neg. incl. aliasing	benign	0.075 / 2/9 / 0.997 / 0.43	0.047 / 4/11 / 0.984 / 0.75	0.010 / 0/3 / 0.043 / 0.10
Conformal LOCO (SAFEVPR protocol)	other cond.	0.052 / 1/9 / 0.997 / 0.34	0.051 / 4/11 / 0.984 / 0.77	0.070 / 2/3 / 0.666 / 0.51
Quantile Transfer	benign	0.097 / 4/9 / 0.998 / 0.47	0.054 / 2/11 / 0.993 / 0.80	0.109 / 2/3 / 0.901 / 0.72
Learned calibrator (CRSG), benign→all	benign	0.099 / 4/9 / 0.996 / 0.33	0.054 / 4/11 / 0.982 / 0.68	0.041 / 0/3 / 0.814 / 0.45
PA ratio, benign→all	benign	0.078 / 2/9 / 0.596 / 0.09	0.046 / 2/11 / 0.449 / 0.19	0.040 / 0/3 / 0.846 / 0.18
Gaussian z-norm null	none	0.350 / 9/9 / 0.999 / 0.72	0.250 / 11/11 / 0.994 / 0.73	0.125 / 3/3 / 0.752 / 0.79
EVT spacing test (F(2,2k))	none	0.058 / 1/9 / 0.888 / 0.17	0.050 / 1/11 / 0.700 / 0.01	0.036 / 0/3 / 0.743 / 0.07
SelfCal on raw max-sim (ablation)	none	0.001 / 0/9 / 0.979 / 0.00	0.010 / 0/11 / 0.949 / 0.46	0.022 / 0/3 / 0.411 / 0.39
Oracle (labels of every condition)	all	0.040 / 0/9 / 0.995 / —	0.049 / 0/11 / 0.992 / —	0.050 / 0/3 / 0.780 / —

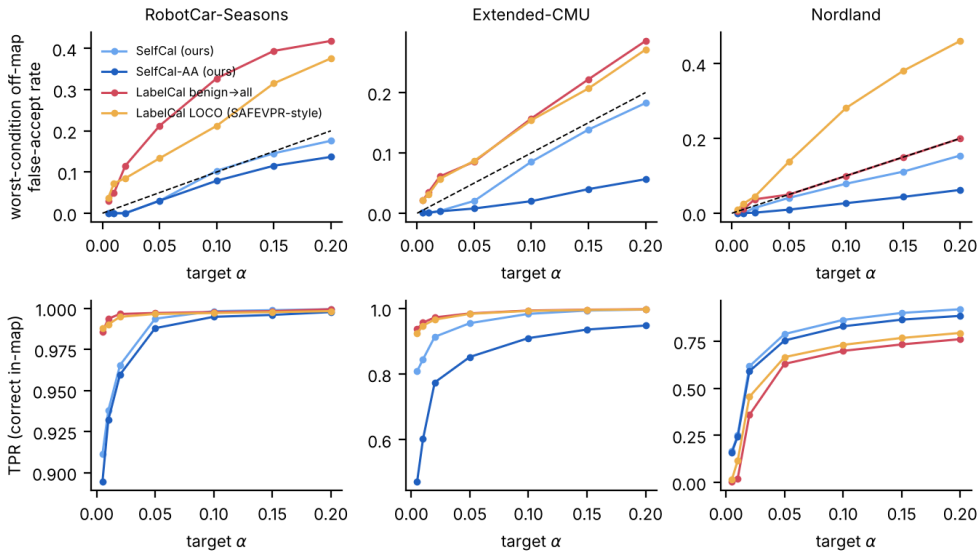


Figure 2. Worst-condition off-map false-accept rate (top) and TPR (bottom) versus the target level, SALAD.

6.3 Why it works

Fig. 1 plots the survival function of the off-map statistic for every RobotCar condition against the map null. All conditions lie below the null for both max-similarity and z_far: appearance change makes unrelated queries **less** map-like than the map's own held-out blocks (T2). Table 3 quantifies the excess-risk bound Δ^+ of T3: with the map null it is at most 0.052 (max-similarity) and 0.047 (z_far, centred) over all 23 conditions, whereas with a labelled benign null it reaches 0.255 (max-similarity, night-rain) — exactly the violations of Table 1. The invariant statistic matters for power: calibrating raw max-similarity on the map is valid but

loses most of its power on Nordland (0.41 vs 0.84), where seasonal change lowers all similarities.

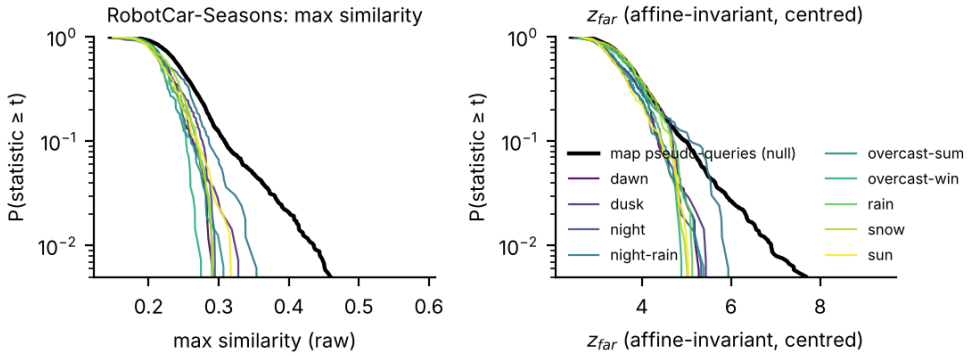


Figure 1. Survival functions of the off-map statistic per RobotCar condition (coloured) vs. the map pseudo-query null (black).

Table 3. Excess-FAR bound Δ⁺_c (one-sided KS) — labelled benign null vs map null

- RobotCar-Seasons (9 cond.): max over conditions — max-sim | benign null 0.255, max-sim | map null 0.017, z_far | benign null 0.146, z_far+centring | map null 0.047
- Extended-CMU (11 dates): max over conditions — max-sim | benign null 0.101, max-sim | map null 0.052, z_far | benign null 0.067, z_far+centring | map null 0.028
- Nordland (3 seasons): max over conditions — max-sim | benign null 0.018, max-sim | map null 0.006, z_far | benign null 0.093, z_far+centring | map null 0.024

6.4 Descriptors

Table 2a (α = 0.05). All 6 descriptors × 3 datasets (18 settings, 138 deployment conditions)

method	labels	violating conditions	settings with ≥1 violation	mean TPR	median aliasing-accept	max worst-cond. FAR
SelfCal-AA (ours)	none	0/138	0/18	0.676	0.008	0.040
SelfCal (ours)	none	0/138	0/18	0.718	0.060	0.051
Threshold, benign→all	benign	25/138	10/18	0.664	0.218	0.281
Threshold, benign, neg. incl. aliasing	benign	15/138	7/18	0.541	0.087	0.209
Conformal LOCO (SAFEVPR protocol)	other cond.	38/138	18/18	0.716	0.231	0.418
Quantile Transfer	benign	64/138	18/18	0.799	0.440	0.483
Learned calibrator (CRSG)	benign	25/138	10/18	0.785	0.173	0.187
PA ratio	benign	11/138	5/18	0.526	0.083	0.122
Gaussian z-norm null	none	112/138	16/18	0.740	0.451	0.525
EVT spacing test	none	4/138	4/18	0.670	0.039	0.109
Oracle (labels of every condition)	all	0/138	0/18	0.721	—	0.050

Table 2 (α = 0.05). Descriptors. Cell: FAR / #viol / TPR / aliasing for SelfCal-AA | #viol for benign threshold, LOCO, CRSG | oracle TPR

descriptor	dataset	R@1 (5 m,10° / 2 fr)	SelfCal-AA	benign thr. #viol (alias)	LOCO #viol	CRSG #viol	oracle TPR
NetVLAD	RobotCar-Season5	0.666	0.006 / 0/9 / 0.746 / 0.00	4/9 (0.07)	2/9	4/9	0.715
NetVLAD	Extended-CMU	0.640	0.006 / 0/11 / 0.594 / 0.01	4/11 (0.20)	4/11	5/11	0.367
NetVLAD	Nordland	0.366	0.009 / 0/3 / 0.413 / 0.02	0/3 (0.07)	1/3	0/3	0.179
CosPlace	RobotCar-Season5	0.767	0.004 / 0/9 / 0.842 / 0.01	0/9 (0.07)	1/9	0/9	0.869
CosPlace	Extended-CMU	0.903	0.012 / 0/11 / 0.813 / 0.01	1/11 (0.24)	3/11	1/11	0.962
CosPlace	Nordland	0.640	0.006 / 0/3 / 0.351 / 0.01	0/3 (0.16)	1/3	0/3	0.393
MixVPR	RobotCar-Season5	0.850	0.004 / 0/9 / 0.912 / 0.00	1/9 (0.04)	1/9	1/9	0.915
MixVPR	Extended-CMU	0.937	0.012 / 0/11 / 0.884 / 0.00	1/11 (0.34)	5/11	1/11	0.940
MixVPR	Nordland	0.748	0.007 / 0/3 / 0.638 / 0.05	0/3 (0.33)	1/3	0/3	0.552
DINOv2 (CLS+mean)	RobotCar-Season5	0.536	0.001 / 0/9 / 0.326 / 0.00	0/9 (0.11)	2/9	0/9	0.642
DINOv2 (CLS+mean)	Extended-CMU	0.637	0.001 / 0/11 / 0.062 / 0.00	2/11 (0.12)	4/11	2/11	0.696
DINOv2 (CLS+mean)	Nordland	0.459	0.007 / 0/3 / 0.183 / 0.01	0/3 (0.06)	1/3	0/3	0.388
SALAD	RobotCar-Season5	0.911	0.004 / 0/9 / 0.988 / 0.00	5/9 (0.50)	1/9	4/9	0.995
SALAD	Extended-CMU	0.936	0.002 / 0/11 / 0.853 / 0.01	4/11 (0.78)	4/11	4/11	0.992
SALAD	Nordland	0.791	0.008 / 0/3 / 0.754 / 0.06	0/3 (0.49)	2/3	0/3	0.780
MegaLoc	RobotCar-Season5	0.905	0.002 / 0/9 / 0.994 / 0.10	1/9 (0.29)	1/9	1/9	0.993
MegaLoc	Extended-CMU	0.927	0.012 / 0/11 / 0.950 / 0.00	2/11 (0.68)	2/11	2/11	0.996
MegaLoc	Nordland	0.799	0.007 / 0/3 / 0.861 / 0.08	0/3 (0.59)	2/3	0/3	0.615

Table 2a aggregates all six descriptors and three datasets (18 settings, 138 deployment conditions) and Table 2 details each setting. SelfCal and SelfCal-AA have **no significant violation in any of the 138 cells** and a worst-condition FAR of at most 0.040 / 0.051 at $\alpha = 0.05$, whereas the benign threshold and the learned calibrator violate in 25 cells (10 of 18 settings), leave-one-condition-out conformal calibration in 38 cells — at least once in **every** setting — and Quantile Transfer in 64. Averaged over all settings, SelfCal reaches the oracle's power (0.718 vs 0.721) and SelfCal-AA trades 4 points of TPR for a 7× lower aliasing acceptance (median 0.008 vs 0.060). For older descriptors (NetVLAD, CosPlace, MixVPR) SelfCal-AA often exceeds the oracle's TPR, because the oracle thresholds raw max-similarity while SelfCal also exploits the far-region statistic. The one weak case is DINOv2 CLS+mean, a generic descriptor that is not trained for VPR: its global features encode the capture condition so strongly that map pseudo-queries (which share the

map's condition) are much more map-like than any deployment query. SelfCal stays valid (FAR ≤ 0.004) but becomes conservative (TPR 0.06–0.33 vs. oracle 0.39–0.70); centring queries on their own session mean recovers only part of the gap (RobotCar 0.33 → 0.41). This is the price of label-free validity, paid only by condition-sensitive descriptors. □

6.5 Batches: FDR and label-free confidence

With an off-map share of 10–50%, Benjamini–Hochberg on SelfCal p-values (q = 0.1) keeps the false-discovery rate of gross errors at 0.004–0.009 on RobotCar and 0.008–0.011 on CMU (max over conditions 0.046 / 0.023) with power 0.96–0.998, and the two-groups confidence has ECE 0.013–0.050 without labels ($\hat{\pi}_0$ within 0.03–0.08 of the true off-map share). On Nordland, where 18.5% of in-map queries are gross aliasing errors that the union rule does not target, BH on the union p-values exceeds q (gross FDR 0.15); with the aliasing-aware p-values, max(p_comb, p_gap), BH keeps the gross FDR at 0.017–0.021 (max 0.030) with power 0.53–0.75. The label-free confidence is, however, poorly calibrated on Nordland (ECE ≈ 0.14): many correct matches on its repetitive scenery have non-significant gaps, which inflates $\hat{\pi}_0$ (0.47–0.91 vs. 0.27–0.57). We therefore recommend the p-values, not the two-groups confidence, on aliasing-prone maps.

Table 5. Deployment batches: BH on SelfCal p-values (SALAD), label-free confidence

- RobotCar-Seasons (9 cond.), off-map share 0.1: π̂0 0.15 (true 0.14); ECE label-free 0.044; BH q=0.1: gross FDR mean 0.004 (max 0.014), power 0.998
- RobotCar-Seasons (9 cond.), off-map share 0.3: π̂0 0.41 (true 0.33); ECE label-free 0.018; BH q=0.1: gross FDR mean 0.009 (max 0.046), power 0.998
- RobotCar-Seasons (9 cond.), off-map share 0.5: π̂0 0.56 (true 0.44); ECE label-free 0.013; BH q=0.1: gross FDR mean 0.004 (max 0.024), power 0.993
- Extended-CMU (11 dates), off-map share 0.1: π̂0 0.11 (true 0.11); ECE label-free 0.050; BH q=0.1: gross FDR mean 0.008 (max 0.018), power 0.972
- Extended-CMU (11 dates), off-map share 0.3: π̂0 0.34 (true 0.31); ECE label-free 0.036; BH q=0.1: gross FDR mean 0.011 (max 0.023), power 0.964
- Extended-CMU (11 dates), off-map share 0.5: π̂0 0.45 (true 0.40); ECE label-free 0.034; BH q=0.1: gross FDR mean 0.011 (max 0.023), power 0.958
- Nordland (3 seasons), off-map share 0.1: π̂0 0.22 (true 0.27); ECE label-free 0.166; BH q=0.1: gross FDR mean 0.144 (max 0.166), power 0.807
- Nordland (3 seasons), off-map share 0.3: π̂0 0.46 (true 0.43); ECE label-free 0.138; BH q=0.1: gross FDR mean 0.146 (max 0.169), power 0.780
- Nordland (3 seasons), off-map share 0.5: π̂0 0.67 (true 0.57); ECE label-free 0.104; BH q=0.1: gross FDR mean 0.147 (max 0.176), power 0.733

6.6 Robustness and ablations

Map in a hard condition. Using Nordland **winter** as the map (queries summer/spring/fall) — so that map pseudo-queries are the degraded ones — SelfCal-AA keeps 0/3 violations (FAR 0.010, aliasing 0.022, TPR 0.593). **Map size.** Keeping 25% of the RobotCar map (635 × 592 m, 1,739 images, 650 pseudo-queries) gives FAR 0.000 and TPR 0.994; 50% gives 0.037 and 0.916. **Hyper-parameters.** Cell size (×0.5–2), exclusion radius (×0.4–2), block size (×0.5–2), δ (×0.6–2) and calibration-set size (497–4,747 pseudo-queries) change TPR by at most 0.05 and keep the worst-condition FAR at or below 0.073 at α = 0.05 (Table 4). **Centring** is essential on Nordland (TPR 0.754 → 0.329 without it) and neutral elsewhere.

Table 4. Ablations of SelfCal-AA (SALAD, α = 0.05): FAR / worst / TPR / aliasing

RobotCar-Seasons (9 cond.)

variant	FAR	worst	TPR	aliasing	null n
default	0.004	0.030	0.988	0.000	2391
cell x0.5	0.004	0.030	0.987	0.000	2391
cell x2.0	0.005	0.036	0.988	0.000	2391
excl x0.4	0.003	0.012	0.941	0.000	2391
excl x2.0	0.004	0.030	0.990	0.000	2391
block_m x0.5	0.001	0.006	0.970	0.000	1384

variant	FAR	worst	TPR	aliasing	null n
block_m x2.0	0.004	0.030	0.991	0.000	2951
off_d_m x0.6	0.001	0.006	0.964	0.000	2851
off_d_m x2.0	0.012	0.073	0.993	0.026	1572
stride 1	0.004	0.030	0.987	0.000	4747
stride 4	0.004	0.030	0.988	0.000	1194
stride 8	0.003	0.024	0.986	0.000	597
no centering	0.002	0.007	0.980	0.000	2391

Nordland (3 seasons)

variant	FAR	worst	TPR	aliasing	null n
default	0.008	0.010	0.754	0.057	2030
cell x0.5	0.005	0.006	0.719	0.049	2030
cell x2.0	0.010	0.011	0.761	0.057	2030
excl x0.4	0.008	0.009	0.748	0.047	2030
excl x2.0	0.008	0.009	0.759	0.076	2030
block_m x0.5	0.001	0.001	0.566	0.044	1810
block_m x2.0	0.011	0.012	0.774	0.060	2140
off_d_m x0.6	0.008	0.009	0.752	0.057	2118
off_d_m x2.0	0.008	0.010	0.758	0.057	1810
stride 1	0.008	0.009	0.753	0.058	4060
stride 4	0.007	0.008	0.750	0.055	1015
stride 8	0.006	0.007	0.747	0.053	497
no centering	0.005	0.006	0.329	0.086	2030

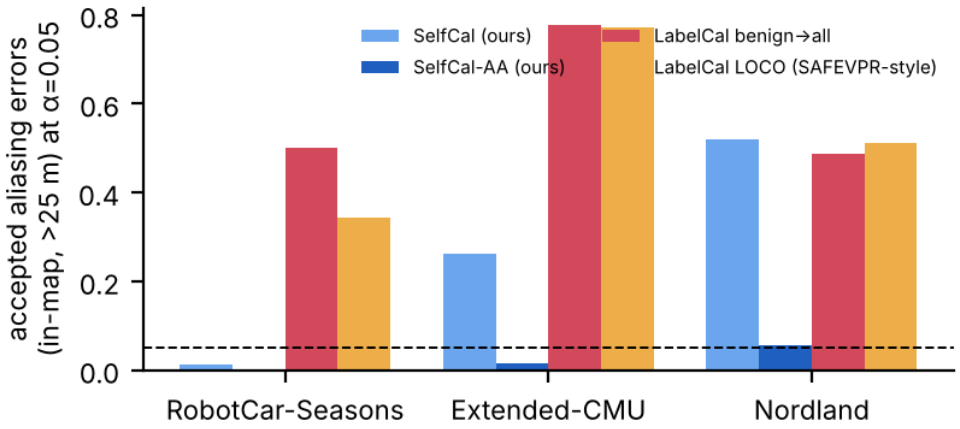


Figure 3. Acceptance rate of in-map aliasing errors (>25 m) at α = 0.05.

7. Limitations

The aliasing guard is supported empirically, not by a theorem. The guarantees concern the **off-map** null and assume deployment off-map queries are no more map-like than the map's own held-out blocks; a deployment area that is a near-duplicate of a mapped area (e.g. two identical stations) is aliasing in disguise. The map must be large enough to hold out blocks (we merged the CMU slices into one city map). RobotCar ground truth exists only for the public training queries (test poses are withheld by the benchmark), so we apply the official thresholds to the top-1 reference pose of the 1,906 training queries. Strict levels ($\alpha \leq 0.01$) cost power on Extended-CMU.

8. Conclusion

A VPR map is more than a retrieval database: with its poses, it is a free, unlimited source of labelled off-map queries in its own environment. Calibrating affine-invariant, geometry-aware statistics against these pseudo-queries gives acceptance decisions whose false-accept rate is controlled without labels and survives the appearance changes that break label-based calibration, together with an aliasing guard, FDR control and calibrated confidence. Natural next steps are sequential evidence (accumulating p-values along a trajectory), maps that change over time, and combining SelfCal with geometric verification scores, which it can calibrate without modification.

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